Exploring root lengths

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- Step 1: Count the observations
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Describe center and spread using invented root lengths of separate seedlings measured once in mm. These practice data teach spreadsheet skills and provide no evidence about real plants.

Part 1: Explore the observations

- **Worksheet:** Roots

Step 1: Inspect the supplied data

- **Read:** Each number is one invented root length in millimeters.
- **Select:** `A2:A6`
- **Compare:** Find the shortest and longest root lengths. Both describe individual roots.
- **Explain:** Tell a classmate what one row represents and what the unit means.
- **Read:** Summary headings are in `D1:E1`; the three Value cells start empty.

Part 2: Summarize root lengths

- **Worksheet:** Roots

Step 1: Count the observations

- **Read:** Sample size is the number of numerical observations, not their sum.
- **Calculate:** In cell E2, enter the formula below.
  In E2, enter:

  ```
  =COUNT(A2:A6)
  ```

- **Check:** Compare the result with the number of supplied root lengths.

Step 2: Calculate the mean

- **Read:** The arithmetic mean adds the root lengths and divides by the sample size.
- **Calculate:** In cell E3, enter the formula below.

In E3, enter:

```
=AVERAGE(A2:A6)
```

- **Explain:** The mean describes the center of these observations in millimeters.

Step 3: Describe the spread

- **Read:** Sample standard deviation describes spread around the mean in millimeters. Use `STDEV.S`; its calculation uses the number of observations minus one.
- **Calculate:** In cell E4, enter the formula below.
  In E4, enter:

  ```
  =STDEV.S(A2:A6)
  ```

- **Check:** Display three decimal places; keep full stored precision.
- **Explain:** Describe mean and SD together: center and spread among root lengths. SD is neither sample size nor the accuracy of an individual measurement.